Saisree Pothu

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Summary

Analytics professional with **6+ years of experience** using product, customer, and financial data to improve conversion, operational efficiency, and business decisions across fintech and education. Experienced in **SQL, Python, ETL, data-quality analysis, automated reporting, dashboards, and root-cause analysis**, including pipelines and roll-up tables that cut processing time by **50%+** and improved pipeline efficiency by **40%**. Built self-service dashboards for **170M+ daily payment transactions**, reducing ad hoc data requests by **70%**, and partners effectively with Product, Engineering, Finance, Risk, and Operations stakeholders to translate complex analyses into actionable insights. Applies AI-enabled analytics with validation and human review.

Experience

Harvard University | *AI Researcher Analyst* **Mar 2026 - Present**

- Deployed fine-tuned models on **AWS EC2** (CPU/GPU instances) and managed evaluation outputs in **Amazon S3**, enabling scalable batch processing and downstream analysis.
- Fine-tuned detection models on **10K+ annotated images to automate eye-misalignment measurement** for clinicians, improving small-object mAP by 18% and reducing missed detections by 24%.
- Built a repeatable **Python pipeline** for scoring, resumable analysis, and **automated reporting**; added data-quality checks to flag unsafe responses and improve the reliability of evaluation results.
- Defined measurement and data quality metrics for assistive navigation models; evaluated **15k real-world queries across scenes** to compare model behavior and identify user-safety gaps.
- Traced performance changes through **error analysis** and controlled comparisons; improved navigation consistency from **23.0% to 63.2%** by testing scene descriptions and image zooming.

Fidelity Investments | *Data Analyst* **Jul 2025 - Dec 2025**

- Built automated **ETL pipelines in Python** to ingest portfolio data from **MSCI, Oracle, and AWS S3 into Snowflake**, automating KPI and risk reporting for **40+ portfolios**, saving **20 team hours per week** and reducing benchmark risk-estimation time by **30%**.
- Used **regression and correlation analysis** to investigate portfolio-to-benchmark alignment, explain risk movement, and improve analytical accuracy by **80%**.
- Built an **agentic risk dashboard** using **Python, SQL, REST APIs, and policy-aware retrieval** to surface exposure changes and deliver AI-written narratives through dashboards and PDF reports.
- Cut estimated report time-to-first-draft from about **6 hours to 15 minutes**; retained analyst review for material changes and added data-quality checks for reconciliations, position completeness, stale data, and limit exceptions.

Cambridge University Press and Assessment | *Senior Data Analyst* **Aug 2023 - Jul 2024**

- Engineered automated **ETL pipelines** using **Python** to ingest data from **PostgreSQL and GCP into BigQuery**, normalizing unstructured **JSON** to extract hierarchical student activity data; optimized extraction time by **80%** and reduced cloud costs via partitioning and indexing, improving query performance by **22%**.
- Automated recurring reporting with **SQL, Python, and Excel**, and built **Tableau dashboards** that reduced ad hoc data requests by **35%**; analyzed engagement across **50+ courses** to isolate dropout drivers, contributing to a **17% revenue increase** and a **23% reduction in student dropout**.
- Developed an **XGBoost-based early-warning model** (**0.84 ROC-AUC**) and dashboards for **1,800+ at-risk learners** to detect disengagement; helped instructors prioritize outreach, reducing manual review by **12 hours monthly** and increasing intervention-cohort completion from **64% to 72%**.
- Led a four-person team building an **AI study coach** for a **200,000-learner platform**; established evaluation measures and quality checks that improved citation relevance from **78% to 93%** and cut unsupported answers from **14.2% to 3.9%**.

Juspay Technologies | *Data Scientist / Data Analyst* **Nov 2019 - Jul 2023**

- Built self-service funnel and performance dashboards using **SQL, Python, and R Shiny**, backed by **Amazon Redshift** for **170M+ daily payment transactions**, reducing ad hoc data requests by **70%** and empowering product and merchant teams.
- Analyzed payment-page **A/B test** results and user funnels to identify conversion friction, driving a **20% increase in transaction rate** and a **30% lift in tokenization**.
- Evaluated payment-routing and retry strategies using historical data to improve payment success by **1.1%**, reduce technical declines by **12%**, and recover **13% more failed transactions**.
- Optimized reporting performance by building automated **ETL pipelines using Python on EC2** and staging raw data in **Amazon S3**, cutting processing time by **50%+** and improving pipeline efficiency by **40%**.
- Automated settlement reconciliation across gateway, bank, and payout data by querying **Amazon DynamoDB** for real-time transaction states, reaching **95% automated coverage** and reducing manual exceptions by **52%**.

Microsoft | *Data Engage Intern* **May 2019 - Jul 2019**

- Built a meeting-time recommendation tool using participant availability and preferences, improving scheduling efficiency by **29%**.

SKILLS

- **Analytics**: SQL analysis, KPI and metric design, hypothesis testing, regression, forecasting, root-cause analysis, data storytelling, Statistical Analysis
- **Programming and data**: SQL, Python, R, PySpark, pandas, NumPy, Spark, BigQuery, PostgreSQL, MySQL, JSON/API data, ETL, Airflow, dbt, Oracle, Redshift, NoSQL, Data Modeling, Data Warehousing, SAS, MATLAB, DynamoDB
- **Reporting and visualization**: Tableau, Power BI, R Shiny, Looker, Excel, Matplotlib, Seaborn, dashboard development, automated reporting, Jira, Amazon QuickSight
- **Machine learning and AI**: XGBoost, Scikit-learn, Predictive Modeling, Classification, Regression Analysis, Model Evaluation (ROC-AUC, mAP), Deep Learning (Fine-tuning), LLM Evaluation
- **Platforms and tools**: AWS, GCP, Databricks, Git

EDUCATION

Northeastern University | *M.S., Data Analytics Engineering* (GPA: 3.9) **Boston, MA Sep 2024 - May 2026**

Lovely Professional University | *B.Tech., Computer Science and Engineering* **India Aug 2016 - May 2020**